

Lista 1 → 2 G.A. Gabriela Vinha Teixeira

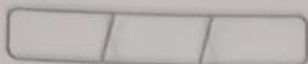
1-a) $\begin{vmatrix} 1 & 2 \\ -4 & 3 \end{vmatrix}$ $\text{Det} = 2 \cdot C_{12} + 3 \cdot C_{22}$
 $\text{Det} = 2 \cdot (-1)^{1+2} \cdot D_{12} + 3 \cdot (-1)^{2+2} \cdot D_{22}$
 $\text{Det} = 2 \cdot (-D_{12}) + 3 \cdot D_{22}$
 $\text{Det} = 2 \cdot (-4) + 3 \cdot 1$
 $\text{Det} = 8 + 3 \Rightarrow \text{Det} = 11$

b) $\begin{vmatrix} \sqrt{2} & 3\sqrt{6} \\ 2 & \sqrt{3} \end{vmatrix}$ $\text{Det} = 3\sqrt{6} \cdot C_{12} + \sqrt{3} \cdot C_{22}$
 $\text{Det} = 3\sqrt{6} \cdot (-1)^{1+2} \cdot D_{12} + \sqrt{3} \cdot (-1)^{2+2} \cdot D_{22}$
 $\text{Det} = 3\sqrt{6} \cdot (-D_{12}) + \sqrt{3} \cdot D_{22}$
 $\text{Det} = 3\sqrt{6} \cdot (-2) + \sqrt{3} \cdot \sqrt{2}$
 $\text{Det} = -6\sqrt{6} + \sqrt{6}$
 $\text{Det} = -5\sqrt{6}$

c) $\begin{vmatrix} \pi & 0 & 2 \\ 5 & -1 & 1 \\ 1 & 0 & 0 \end{vmatrix}$ $\text{Det} = 1 \cdot C_{31} + 0 \cdot C_{32} + 0 \cdot C_{33}$
 $\text{Det} = 1 \cdot (-1)^{3+1} \cdot D_{31}$
 $\text{Det} = + D_{31} \Rightarrow \begin{vmatrix} 0 & 2 \\ -1 & 1 \end{vmatrix} = 2$
 $\text{Det} = 2$

d) $\begin{vmatrix} -2 & 1 & -1 \\ 1 & 5 & 4 \\ -3 & 4 & 2 \end{vmatrix}$ $\text{Det} = -2 \cdot C_{11} + 1 \cdot C_{12} + (-1) \cdot C_{13}$
 $\text{Det} = -2 \cdot (-1)^{1+1} \cdot D_{11} + 1 \cdot (-1)^{1+2} \cdot D_{12} + (-1) \cdot (-1)^{1+3} \cdot D_{13}$
 $\text{Det} = -2 \cdot \begin{vmatrix} 5 & 4 \\ 4 & 2 \end{vmatrix} - \begin{vmatrix} 1 & 4 \\ -3 & 2 \end{vmatrix} + \begin{vmatrix} 1 & 5 \\ -3 & 4 \end{vmatrix}$
 $\text{Det} = -2 + (-16 + 16) - (12 + 2) - (15 + 4)$
 $\text{Det} = -2 - 6 - 19 - 19$
 $\text{Det} = 12 - 19 - 19 = -26$

e) $\begin{vmatrix} 0 & 2 & 0 \\ 1 & 3 & 5 \\ 2 & -1 & 2 \end{vmatrix}$ $\text{Det} = 0 \cdot C_{11} + 2 \cdot C_{12} + 0 \cdot C_{13}$
 $\text{Det} = 2 \cdot (-1)^{1+2} \cdot D_{12}$
 $\text{Det} = -2 \cdot \begin{vmatrix} 1 & 5 \\ 2 & 2 \end{vmatrix}$
 $\text{Det} = -2 \cdot (-8) = 16$
 $\text{Det} = 16$



$$\begin{array}{l|l} \text{f)} & \begin{array}{cccc|l} 3 & -1 & 1 & 1 & \text{Det} = 3 \cdot C_{11} + 0 \cdot C_{21} + 0 \cdot C_{31} + 0 \cdot C_{41} \\ 0 & 1 & 0 & 1 & \text{Det} = 3 - (-1)^{1+1} \cdot D_{11} \\ 0 & -1 & 1 & 0 & \text{Det} = 3 \cdot \begin{vmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} \\ 0 & 0 & 1 & -1 & \end{array} \end{array}$$

$$\text{Det} = 3 \cdot (-2)$$

$$\text{Det} = -6$$

$$\begin{array}{l|l} \text{g)} & \begin{array}{cccccc|l} 1 & 0 & 0 & 0 & 0 & \text{Det} = 1 \cdot C_{11} + 0 \cdot C_{12} + 0 \cdot C_{13} + 0 \cdot C_{14} + 0 \cdot C_{15} \\ 5 & 1 & 2 & 5 & 3 & \text{Det} = 1 \cdot (-1)^{1+1} \cdot D_{11} \\ 7 & 2 & \sqrt{5} & 0 & 0 & \text{Det} = 1 \cdot \begin{vmatrix} 1 & 2 & 5 & 3 \\ 2 & \sqrt{5} & 0 & 0 \\ -3 & 6 & 1 & 0 \\ -3 & 0 & 0 & 0 \end{vmatrix} \\ 10 & -3 & 6 & 1 & 0 & \\ -2 & -3 & 0 & 0 & 0 & \end{array} \end{array}$$

$$\text{Det} = 1 \cdot 3 \cdot (-1)^{1+4} \cdot D_{14}$$

$$\text{Det} = 1 \cdot (-3) \cdot \begin{vmatrix} 2 & \sqrt{5} & 0 \\ -3 & 6 & 1 \\ -3 & 0 & 0 \end{vmatrix}$$

$$\text{Det} = -3 \cdot -3\sqrt{5}$$

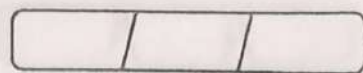
$$\text{Det} = 9\sqrt{5}$$

$$\begin{array}{l|l} \text{h)} & \begin{array}{cccccc|l} 3 & 0 & 0 & 0 & 0 & \text{Det} = 0 \cdot C_{41} + 0 \cdot C_{42} + 0 \cdot C_{43} + 1 \cdot C_{44} + 0 \cdot C_{45} \\ 0 & 0 & 0 & 0 & -2 & \text{Det} = C_{44} \\ 0 & 0 & 2 & 0 & 0 & \text{Det} = (-1)^{4+4} \cdot D_{44} \\ 0 & 0 & 0 & 1 & 0 & \text{Det} = 1 \cdot \begin{vmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & -2 & 0 & 0 \end{vmatrix} \\ 0 & -2 & 0 & 0 & 0 & \end{array} \end{array}$$

$$\text{Det} = 1 \cdot 3 \cdot (-1)^{4+4} \cdot \begin{vmatrix} 0 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 0 \end{vmatrix}$$

$$\text{Det} = 1 \cdot 3 \cdot (-8)$$

$$\text{Det} = -24$$



2-a) $\text{Det}(A+B) \Rightarrow A = \begin{pmatrix} 3 & -5 & 7 \\ 4 & 2 & 8 \\ 1 & -9 & 6 \end{pmatrix} + B = \begin{pmatrix} 4 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -9 \end{pmatrix}$

$$\text{Det}(A+B) = \begin{vmatrix} 7 & -2 & 14 \\ 3 & 2 & 10 \\ 4 & -8 & 2 \end{vmatrix}$$

$$= -112 + 560 + 12 + 28 - 336 - 80$$

$$\text{Det}(A+B) = 72$$

b) $\text{Det}(A \cdot B) \Rightarrow$ Sabiendo que $\text{Det}(A \cdot B) = \text{Det} A \cdot \text{Det} B$

$$\text{Det}(A \cdot B) = \begin{vmatrix} 3 & -5 & 7 \\ 4 & 2 & 8 \\ 1 & -9 & 6 \end{vmatrix} \cdot \begin{vmatrix} 4 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -9 \end{vmatrix}$$

$$\text{Det}(A \cdot B) = (36 - 252 - 90 - 14 + 216 + 120) \cdot (-2 + 18 + 8 - 12)$$

$$\text{Det}(A \cdot B) = 66 \cdot 7$$

$$\text{Det}(A \cdot B) = 462$$

c) $\text{det}(B^t A^t)$; Sabiendo que $\text{Det}(B^t A^t) = \text{Det}(B^t) \cdot \text{Det}(A^t)$ e
también, $\text{Det}(A^t) = -\text{Det}(A)$ e $\text{Det}(B^t) = -\text{Det}(B)$

$$\text{Det}(B^t A^t) = -7 \cdot -66$$

$$\text{Det}(B^t A^t) = 462$$

d) $\text{det}(3A - 2C + B)$

$$3 \cdot \begin{pmatrix} 3 & -5 & 7 \\ 4 & 2 & 8 \\ 1 & -9 & 6 \end{pmatrix} - 2 \cdot \begin{pmatrix} 2 & 3 & -1 \\ 6 & 9 & -2 \\ 8 & 12 & -3 \end{pmatrix} + \begin{pmatrix} 4 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -9 \end{pmatrix}$$

$$\begin{pmatrix} 9 & -15 & 21 \\ 12 & 6 & 24 \\ 3 & -27 & 18 \end{pmatrix} - \begin{pmatrix} 4 & 6 & -2 \\ 12 & 18 & -4 \\ 16 & 24 & -6 \end{pmatrix} + \begin{pmatrix} 4 & 3 & 7 \\ -1 & 0 & 2 \\ 3 & 1 & -9 \end{pmatrix}$$

$$\begin{pmatrix} 9 & -18 & 30 \\ -1 & -12 & 30 \\ -10 & -50 & 20 \end{pmatrix} \Rightarrow \text{Det}$$

$$\text{Det} = \begin{vmatrix} 9 & -18 & 30 \\ -1 & -12 & 30 \\ -10 & -50 & 20 \end{vmatrix} \Rightarrow \begin{vmatrix} 9 & -18 & 3 \\ -1 & -12 & 3 \\ -10 & -50 & 2 \end{vmatrix} \cdot 10$$

$$\text{Det} = 9 \cdot (-12) \cdot 2 + (-18) \cdot 3 \cdot (-10) + 30 \cdot (-1) \cdot (-50) - ((-10) \cdot (-12) \cdot 3 + (-50) \cdot 3 \cdot 9 + 2 \cdot (-1) \cdot (-18)) \cdot 10$$

$$\text{Det} = -216 + 540 + 150 - (360 + 1350 + 36) \cdot 10$$

$$\text{Det} = 474 + 954 \cdot 10$$

$$\text{Det} = 1428 \cdot 10$$

$$\text{Det} = 14280$$

e) $\text{det}(A^t)$, Sabiendo que $\text{det}(AC^t) = \text{det} A \cdot -\text{det} C$

$$\text{det}(AC^t) = \text{Det} A \cdot -\text{Det} C \times 1,5$$

$$66 \cdot - \begin{vmatrix} 2 & 3 & -1 \\ 6 & 9 & -2 \\ 8 & 12 & -3 \end{vmatrix}$$

$$66 \cdot -54 - 98 - 72 - (-72 - 98 - 54)$$

$$66$$

$$0$$

$$\text{det}(A^t) = 0$$

3- Sabiendo que $\text{det}(A) = -2$

a) $\text{Det}(A^t) = -\text{Det}(A) \cdot$ b) $\text{Det}(6 \cdot A) = 6^4 \cdot -2$

$$\text{Det}(A^t) = -(-2) \Rightarrow \text{Det}(6A) = 1296 \cdot (-2)$$

$$\text{Det}(A^t) = 2$$

$$\text{Det}(6A) = -2592$$

c) $\text{Det}(A^2) = (\text{Det} A)^2$ c) $\text{Det}(A^{-1}) = (\text{Det}(A))^{-1}$

$$\text{Det}(A^2) = (-2)^2$$

$$\text{Det}(A^{-1}) = -\frac{1}{2}$$

$$\text{Det}(A^2) = -128$$

$$\begin{array}{|ccc|} \hline 4-a & a & b & c \\ \hline & d & e & f \\ \hline & 9g & 4h & 4i \end{array} \xrightarrow{\times 4} = 4 \cdot -3 = -12$$

$$\begin{array}{|ccc|} \hline b) & a & b & -2c \\ \hline \times 3 & 3d & 3e & -6f \\ \hline & g & h & -2i \end{array} \xrightarrow{\times 2} = -3 \cdot 3 \cdot -2 = 18$$

$$\begin{array}{|ccc|} \hline c) & -a & -b & -c \\ \hline & g & h & i \\ \hline & -d & -e & -f \end{array} \xrightarrow{\times -1} \begin{array}{l} l_3 \rightarrow l_2 \\ = 3 \cdot -1 \cdot -1 \end{array}$$

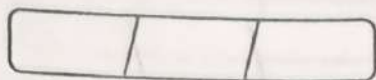
$$\begin{array}{|ccc|} \hline d) & g & h & i \\ \hline & a & b & c \\ \hline & d & e & f \end{array} \xrightarrow{\text{"2 trocar de linhas"}} = -3$$

$$\begin{array}{|ccc|} \hline e) & a & b & c \\ \hline & 2d+a & 2e+b & 2f+c \\ \hline & f & h & i \end{array} = 2 \cdot (-3) \cdot -6$$

$$\begin{array}{|ccc|} \hline f) & ka+a & kb+b & ck+c \\ \hline & d & e & f \\ \hline & g & h & i \end{array} \begin{array}{l} (k+1)a \quad (k+1)b \quad (k+1)c \\ \det = -3 \cdot k+1 \rightarrow -3k-3 \end{array}$$

$$\begin{array}{|cccc|} \hline 5- & 10 & 8 & 90 & -2 \\ \hline A = & 4 & 6 & 20 & -4 \\ \hline & -5 & -7 & -30 & 1 \\ \hline & 6 & 6 & -30 & 12 \end{array} \Rightarrow \begin{array}{|cccc|} \hline & 10 & 8 & 4 & -2 \\ \hline & 4 & 6 & 2 & -4 \\ \hline & -5 & -7 & -3 & 1 \\ \hline & 6 & 6 & -3 & 12 \end{array} \begin{array}{l} \rightarrow \div 2 \\ \cdot 10 \\ \rightarrow \div 2 \\ \rightarrow \div 3 \end{array}$$

$$\begin{array}{|cccc|} \hline = & 5 & 4 & 2 & -1 \\ \hline & 2 & 3 & 1 & -2 \\ \hline & -5 & -7 & -3 & 1 \\ \hline & 2 & 2 & -1 & 4 \end{array} \cdot 10 \cdot 2 \cdot 2 \cdot 3 \Rightarrow \text{entre folhas}$$



continuação:

5-	5	4	2	-1	
	2	3	1	-2	• 120
	-5	-7	-3	1	
	2	2	-1	4	

	23-2		54-1		54-1		54-1
2 · (-1) ⁴⁺³	-5-7	+ 1 · (-1) ³⁺²	-5-7	+ (-5) · (-1) ³⁺³	23-2	+ (-1) · (-1) ³⁺⁴	23-2
	224		224		224		-5-7

$$2(-56+20+6-28-4+60) - 1(-140+8+10-14-60+80) - 3(60-4-16+6+20-32) + (15+90+14-15-70-9)$$

$$2(-2) - 1(-66) - 3(34) + 1(-24)$$

$$(-4) + 66 - 102 - 24$$

$$66 - 130$$

$$-64 \quad \bullet \quad 120$$

$$- 7680$$

6-a)	4	6	x		4	3	x	4	3
	7	4	2x	= -128 =>	7	2	2x	7	2
	5	2	-x		5	1	-x	5	1

÷ 2

$$-8x + 30x + 7x - 10x - 8x + 21x$$

$$58x - 26x$$

$$32x = -64$$

$$x = -2$$

b)	3	5	7	3	5
	2x	x	3x	2x	x
	4	6	7	4	6

$$= 39$$

$$21x + 60x + 84x - 70x - 54x - 28x = 39$$

$$165x - 152x = 39$$

$$13x = 39$$

$$x = 3$$

c)	$x+3$	$x+1$	$x+4$	$x+3$	$x+1$	
	4	5	3	4	5	$= -7$
	9	10	7	9	10	

$$35x + 105 + 27x + 27 + 40x + 160 - 45x - 180 - 30x - 90 - 28x - 28$$

$$102x + 292 - 103x - 298 = -7$$

$$-x - 6 = -7$$

$$-x = -1$$

$$x = 1$$

d) $\begin{pmatrix} x & x+2 \\ 1 & x \end{pmatrix}$ e' singular

$$\begin{vmatrix} x & x+2 \\ 1 & x \end{vmatrix} = 0 \Rightarrow x^2 - x - 2 = 0$$

$$\Delta = 1^2 - 4 \cdot (-2) \cdot 1 = 9$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2 \cdot a} \Rightarrow \frac{1 \pm 3}{2}$$

$$\frac{1+3}{2} = \frac{4}{2} = 2$$

$$\frac{1-3}{2} = \frac{-2}{2} = -1$$

$$\therefore S = \{-1, 2\}$$

e) $\begin{vmatrix} x-4 & 0 & 3 \\ 2 & 0 & x-9 \\ 0 & 3 & 0 \end{vmatrix}$ e' invertible $\rightarrow \det \neq 0$

$x-4$	0	3	$x-4$	0
2	0	$x-9$	2	0
0	3	0	0	3

$$0 + 0 + 18 - (0 + (3x - 27) \cdot (x - 9) + 0)$$

$$18 - (3x^2 - 12x - 27x + 108)$$

$$-3x^2 + 39x - 90 \neq 0 \quad \div (-3)$$

$$x^2 - 13x + 30 \neq 0$$

$$\Delta = 13^2 - 4 \cdot 30 \cdot 1$$

$$\Delta = 169 - 120$$

$$\Delta = 49$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} \rightarrow \frac{13 \pm \sqrt{49}}{2}$$

$$\frac{13+7}{2} = \frac{20}{2} = 10$$

$$\frac{13-7}{2} = \frac{6}{2} = 3$$

$$S = \{x \in \mathbb{R} / x \neq 10 \text{ ou } x \neq 3\}$$

$$7- A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$a) A^{-1} \rightarrow A A^{-1} = A^{-1} A = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

A^{-1} existe somente se $\det(A) \neq 0$, então
 $ad - bc \neq 0$

$$b) A = \begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} \rightarrow \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$$

$$6 - 5 = \det(A) = 1$$

$$B = \begin{vmatrix} 4 & 7 \\ 1 & 2 \end{vmatrix} \rightarrow \begin{pmatrix} 2 & -7 \\ -1 & 4 \end{pmatrix}$$

$$8 - 7 = 1 \rightarrow \det(B)$$

$$(AB)^{-1} = B^{-1} \cdot A^{-1}$$

$$\begin{pmatrix} 2 & -7 \\ -1 & 4 \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} = \begin{pmatrix} 2 \cdot 2 + (-7) \cdot (-5) & 2 \cdot (-1) + (-7) \cdot 3 \\ -1 \cdot 2 + 4 \cdot (-5) & -1 \cdot (-1) + 4 \cdot 3 \end{pmatrix} =$$

$$= \begin{pmatrix} 4 + 35 & -2 - 21 \\ -2 - 20 & 1 + 12 \end{pmatrix} \Rightarrow (AB)^{-1} = \begin{pmatrix} 39 & -23 \\ -22 & 13 \end{pmatrix}$$

8 - a) $A = \begin{pmatrix} 2 & -2 \\ 3 & 1 \end{pmatrix} \rightarrow$ cofactores:

$$C_{11} = (-1)^{1+1} \cdot 1 = 1$$

$$C_{12} = (-1)^{1+2} \cdot 3 = -3$$

$$CA = \begin{pmatrix} 1 & -3 \\ 2 & 2 \end{pmatrix}$$

$$C_{21} = (-1)^{2+1} \cdot -2 = 2$$

$$C_{22} = (-1)^{2+2} \cdot 2 = 2$$

b) $B = \begin{pmatrix} 2 & -2 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$

$$C_{11} = (-1)^{1+1} \cdot \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

$$C_{12} = (-1)^{1+2} \cdot \begin{vmatrix} 1 & 1 \\ 0 & -1 \end{vmatrix} = 1$$

$$C_{13} = (-1)^{1+3} \cdot \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = -1$$

$$C_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -2 & 0 \\ 1 & -1 \end{vmatrix} = -2$$

$$C_{31} = (-1)^{3+1} \cdot \begin{vmatrix} -2 & 0 \\ 2 & 1 \end{vmatrix} = -2$$

$$C_{22} = (-1)^{2+2} \cdot \begin{vmatrix} 2 & 0 \\ 0 & -1 \end{vmatrix} = -2$$

$$C_{32} = (-1)^{3+2} \cdot \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = -2$$

$$C_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix} = -2$$

$$C_{33} = (-1)^{3+3} \cdot \begin{vmatrix} 2 & -2 \\ 1 & 2 \end{vmatrix} = 6$$

$$CB = \begin{pmatrix} -3 & 1 & -1 \\ -2 & -2 & -2 \\ -2 & -2 & 6 \end{pmatrix}$$

9. Sabiendo que: $A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$

a) $A = \begin{pmatrix} 2 & -2 \\ 3 & 1 \end{pmatrix}$ $\det: 2 \cdot 1 + (-3 \cdot (-2))$
 $\det: 2 + 6 = 8$

cofactores de A

$$C_{11} = (-1)^{1+1} \cdot 1 = 1 \quad C_{21} = (-1)^{2+1} \cdot (-2) = 2$$

$$C_{12} = (-1)^{1+2} \cdot (-3) = 3 \quad C_{22} = (-1)^{2+2} \cdot (2) = 2$$

$$CA = \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} \rightarrow \text{adj}(A) = \begin{pmatrix} 1 & 2 \\ -3 & 2 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} \frac{1}{8} & \frac{2}{8} \\ -\frac{3}{8} & \frac{2}{8} \end{pmatrix} = \begin{pmatrix} \frac{1}{8} & \frac{1}{4} \\ -\frac{3}{8} & \frac{1}{4} \end{pmatrix}$$

$$b) B = \begin{pmatrix} 2 & -2 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix} \Rightarrow \det(B) = -4 + 0 + 0 - 0 - 2 - 2 \\ \det(B) = -8$$

considerando CB do ex 8, temos

$$CB = \begin{pmatrix} -3 & -1 & 1 \\ 2 & -2 & -2 \\ 1 & -2 & 2 \end{pmatrix} \rightarrow \underset{\substack{\downarrow \\ C^t}}{\text{adj}(B)} = \begin{pmatrix} -3 & 2 & 1 \\ -1 & -2 & -2 \\ 1 & -2 & 2 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} -3 & 2 & 1 \\ -1 & -2 & -2 \\ 1 & -2 & 2 \end{pmatrix} \div -8 \Rightarrow \begin{pmatrix} 3/8 & -1/4 & -1/8 \\ 1/8 & -1/4 & -1/4 \\ -1/8 & 1/4 & 1/4 \end{pmatrix}$$

$$c) C = \begin{pmatrix} 0 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix} \Rightarrow \det(C) = 0 + 2 + 1 - (0 + 0 + 0) \\ \det(C) = 3$$

$$C_{11} = 1 \cdot \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} = 1 \quad C_{21} = -1 \cdot \begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix} = -1 \quad C_{31} = 1 \cdot \begin{vmatrix} -1 & 1 \\ 0 & -1 \end{vmatrix} = 1 \\ C_{12} = -1 \cdot \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} = -1 \quad C_{22} = 1 \cdot \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 \quad C_{32} = -1 \cdot \begin{vmatrix} 0 & 1 \\ 2 & -1 \end{vmatrix} = -2 \\ C_{13} = 1 \cdot \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} = 2 \quad C_{23} = -1 \cdot \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} = -1 \quad C_{33} = 1 \cdot \begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} = 2$$

$$C^t = \begin{pmatrix} 1 & -1 & 2 \\ -1 & -1 & -2 \\ 1 & -2 & -2 \end{pmatrix} \rightarrow \underset{\substack{\downarrow \\ C^t}}{\text{adj}C} = \begin{pmatrix} 1 & -1 & 1 \\ -1 & -1 & -2 \\ 2 & 1 & -2 \end{pmatrix}$$

$$C^{-1} = \begin{pmatrix} 1/3 & -1/3 & 1/3 \\ -1/3 & -1/3 & -2/3 \\ 2/3 & 1/3 & -2/3 \end{pmatrix}$$

$$d) \quad D = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 2 & 3 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad \det(D) = 0 \cdot \begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 3 \\ 0 & -1 & 0 \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 3 \\ 0 & -1 & 0 \end{vmatrix} + 0 \cdot \begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 3 \\ 0 & -1 & 0 \end{vmatrix} + 0 \cdot \begin{vmatrix} 1 & 0 & 1 \\ 2 & 2 & 3 \\ 0 & -1 & 0 \end{vmatrix}$$

$$\det = -1 \cdot -2 - (-3)$$

$$\det = 1$$

$$C_{11} = 3 \quad C_{21} = 2 \quad C_{31} = 1 \quad C_{41} = 0$$

$$C_{12} = 2 \quad C_{22} = 1 \quad C_{32} = 0 \quad C_{42} = 0$$

$$C_{13} = 0 \quad C_{23} = 0 \quad C_{33} = -1 \quad C_{43} = -3$$

$$C_{14} = 2 \quad C_{24} = 0 \quad C_{34} = 1 \quad C_{44} = 2$$

$$C_D = \begin{pmatrix} 3 & 2 & 0 & 2 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & -1 & 1 \\ 0 & 0 & -3 & 2 \end{pmatrix} \rightarrow \text{adj } D = \begin{pmatrix} 3 & 2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & -1 & -3 \\ 2 & 0 & 1 & 2 \end{pmatrix} \Rightarrow D^{-1}$$

$$10^{-1} \cdot 2A^t = C - XB$$

$$a) \quad 2A^t = C - XB$$

$$2A^t + XB = C$$

$$XB = C - 2A^t$$

$$X = \frac{C - 2A^t}{B}$$

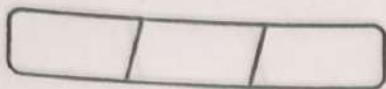
$$X = (C - 2A^t) B^{-1}, \text{ onde } \det(B) \neq 0$$

$$b) \quad X = (C - 2A^t) \cdot B^{-1}$$

$$A^t = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \\ -1 & 0 & 2 \end{pmatrix} \quad B^{-1} = \begin{vmatrix} 3 & -2 & 6 \\ 2 & -1 & 5 \\ 1 & 0 & 3 \end{vmatrix} \rightarrow B^{-1} = \frac{1}{-1} \cdot \text{adj } B = -\text{adj } B$$

$$6 + 12 - 9 - 10 = -1$$

$$CB = \begin{pmatrix} -3 & -1 & 1 \\ 6 & 3 & -2 \\ -4 & -3 & 1 \end{pmatrix} \quad \begin{aligned} C_{11}(-1)^{1+1} \begin{vmatrix} -1 & 5 \\ 0 & 3 \end{vmatrix} &= -3 & C_{21}(-1)^{2+1} \begin{vmatrix} -2 & 6 \\ 0 & 3 \end{vmatrix} &= 6 & C_{31}(-1)^{3+1} \begin{vmatrix} -2 & 6 \\ -1 & 5 \end{vmatrix} &= 4 \\ C_{12}(-1)^{1+2} \begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} &= -1 & C_{22}(-1)^{2+2} \begin{vmatrix} 3 & 6 \\ 1 & 3 \end{vmatrix} &= 3 & C_{32}(-1)^{3+2} \begin{vmatrix} 3 & 6 \\ -1 & 3 \end{vmatrix} &= -3 \\ C_{13}(-1)^{1+3} \begin{vmatrix} -2 & 6 \\ 0 & 3 \end{vmatrix} &= 1 & C_{23}(-1)^{2+3} \begin{vmatrix} 3 & -2 \\ 1 & 0 \end{vmatrix} &= -2 & C_{33}(-1)^{3+3} \begin{vmatrix} 3 & -2 \\ -1 & 3 \end{vmatrix} &= 1 \end{aligned}$$



Sabendo que:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \\ -1 & 0 & 2 \end{pmatrix}$$

$$\text{adj}(B) = \begin{pmatrix} -3 & 6 & -4 \\ -1 & 3 & -3 \\ 1 & -2 & 1 \end{pmatrix}$$

$$B^{-1} = -\text{adj}$$

$$\Rightarrow B^{-1} = \begin{pmatrix} 3 & -6 & 4 \\ 1 & -3 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

$$X = (C - 2A^t)B^{-1}$$

$$X = \left(\begin{pmatrix} 4 & 4 & 0 \\ 0 & 8 & -2 \\ -2 & 0 & 6 \end{pmatrix} - 2 \cdot \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \\ -1 & 0 & 2 \end{pmatrix} \right) \cdot \begin{pmatrix} 3 & -6 & 4 \\ 1 & -3 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

$$X = \left(\begin{pmatrix} 4 & 4 & 0 \\ 0 & 8 & -2 \\ -2 & 0 & 6 \end{pmatrix} - \begin{pmatrix} 2 & 4 & 0 \\ 0 & 6 & -2 \\ -2 & 0 & 4 \end{pmatrix} \right) \cdot \begin{pmatrix} 3 & -6 & 4 \\ 1 & -3 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

$$X = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 3 & -6 & 4 \\ 1 & -3 & 3 \\ -1 & 2 & -1 \end{pmatrix}$$

$$X = \begin{pmatrix} 2 \cdot 3 + 0 \cdot 1 + 0 \cdot (-1) & 2 \cdot (-6) + 0 \cdot (-3) + 0 \cdot 2 & 2 \cdot 4 + 0 \cdot 3 + 0 \cdot (-1) \\ 0 \cdot 3 + 2 \cdot 1 + 0 \cdot (-1) & 0 \cdot (-6) + 2 \cdot (-3) + 0 \cdot 2 & 0 \cdot 4 + 2 \cdot 3 + 0 \cdot (-1) \\ 0 \cdot 3 + 0 \cdot 1 + 2 \cdot (-1) & 0 \cdot (-6) + 0 \cdot (-3) + 2 \cdot 2 & 0 \cdot 4 + 0 \cdot 3 + 2 \cdot (-1) \end{pmatrix}$$

$$X = \begin{pmatrix} 6 & -12 & 8 \\ 2 & -6 & 6 \\ -2 & 4 & -2 \end{pmatrix}$$